

LOM, Bulgaria

WMO: 155110

Lat: 43.817N

Lon: 23.250E

Elev: 108

StdP: 14.64

Time Zone: 2.00 (EUE)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.7	17.4	10.2	9.3	14.0	14.8	11.7	18.9	27.7	44.0	23.4	40.2	3.5	50	0.443

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	92.4	75.7	89.5	74.7	86.9	73.6	78.0	88.5	76.3	86.8	74.7	84.8	4.8	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.6	130.1	85.2	72.8	122.3	83.2	71.3	116.0	81.5	41.4	88.8	39.8	87.0	38.2	84.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
26.3	21.7	17.9	DB	6.6	96.5	6.3	3.4	2.0	98.9	-1.7	100.9	-5.2	102.8	-9.8	105.3
			WB	7.0	81.2	6.3	2.7	2.5	83.2	-1.2	84.8	-4.7	86.3	-9.3	88.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.0	32.9	36.8	45.9	55.5	65.2	72.1	76.1	75.2	66.2	55.1	43.9	34.2	(6)
	DBStd	16.81	8.39	9.37	8.00	5.91	5.68	5.27	4.60	4.86	5.74	7.13	6.95	7.82	(7)
	HDD50	1830	530	374	174	20	0	0	0	0	33	207	492		(8)
	HDD65	4702	995	789	592	287	70	7	1	2	54	316	633	956	(9)
	CDD50	3667	1	5	47	186	472	663	809	781	487	190	24	2	(10)
	CDD65	1061	0	0	0	3	76	220	345	318	91	8	0	0	(11)
	CDH74	8963	0	0	1	30	530	1860	3212	2708	569	52	1	0	(12)
	CDH80	3232	0	0	0	3	89	611	1316	1071	137	5	0	0	(13)
(14) Wind	WSAvg	5.4	5.1	6.0	6.6	5.8	5.7	5.6	5.6	4.9	5.0	4.6	4.8	4.8	(14)
(15) Precipitation	PrecAvg	22.1	1.5	1.5	1.5	1.9	2.4	2.1	2.2	1.7	1.8	1.9	1.5	2.0	(15)
	PrecMax	34.2	3.3	2.9	3.9	3.9	4.5	4.1	5.2	8.2	4.7	5.6	3.7	4.6	(16)
	PrecMin	12.2	0.2	0.1	0.3	0.0	0.8	0.7	0.2	0.1	0.2	0.2	0.0	0.4	(17)
	PrecStd	5.1	0.8	0.9	0.9	0.9	1.0	1.0	1.5	1.7	1.3	1.4	1.0	1.2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.5	62.4	71.1	78.1	86.2	93.6	98.0	96.3	88.1	80.0	67.7	57.8	(19)
		MCWB	49.3	52.0	56.2	63.4	69.8	75.9	75.1	75.3	71.6	66.7	57.5	50.3	(20)
	2%	DB	51.7	57.8	66.5	72.7	82.3	89.2	92.9	92.1	84.1	73.7	61.2	50.0	(21)
		MCWB	46.3	49.7	54.4	60.0	68.4	74.5	75.4	76.2	69.6	64.5	54.9	45.7	(22)
	5%	DB	47.3	53.5	62.5	69.1	79.6	86.4	89.7	89.2	80.2	69.3	56.8	46.0	(23)
		MCWB	42.8	46.9	52.5	58.0	67.3	73.1	74.7	74.9	68.2	62.0	52.9	43.1	(24)
	10%	DB	43.5	49.3	58.5	66.2	76.8	83.5	86.9	86.2	76.9	65.5	53.5	43.5	(25)
		MCWB	40.2	44.2	50.1	56.6	66.1	71.9	74.0	73.4	66.3	59.5	50.5	41.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.1	53.3	57.5	65.2	72.7	78.9	79.7	81.0	73.6	68.4	59.3	50.4	(27)
		MCDB	56.7	61.1	67.7	74.2	81.7	89.4	90.7	90.2	83.6	76.6	65.2	56.5	(28)
	2%	WB	46.1	49.9	55.1	61.7	70.2	76.2	77.9	78.1	71.3	65.3	56.1	46.5	(29)
		MCDB	51.3	57.1	65.0	70.3	79.7	86.8	88.6	88.5	82.0	72.0	60.3	49.5	(30)
	5%	WB	43.0	47.3	52.7	59.3	68.4	74.3	76.3	76.2	69.2	62.7	53.4	43.7	(31)
		MCDB	46.6	53.1	61.3	66.9	77.5	84.4	86.9	86.5	78.1	68.7	56.0	45.6	(32)
	10%	WB	40.6	44.6	50.5	57.5	66.6	72.5	74.8	74.3	67.3	60.2	50.9	41.4	(33)
		MCDB	43.3	48.9	57.6	64.4	75.4	82.1	85.1	84.2	75.1	64.8	53.5	43.1	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.2	11.8	15.2	16.2	17.4	18.2	19.2	19.2	17.3	14.2	9.7	8.3	(35)
		MCDBR	16.1	19.8	23.0	21.7	22.4	22.7	23.7	23.8	22.8	20.5	16.4	12.7	(36)
	5% WB	MCWBR	11.4	13.2	13.4	11.6	11.2	11.3	12.0	12.1	11.7	12.6	10.8	9.2	(37)
		MCWBR	14.8	18.8	21.5	19.4	20.6	21.5	21.8	21.8	21.0	19.4	14.3	11.3	(38)
(40) Clear-Sky Solar Irradiance	taub		0.348	0.377	0.417	0.479	0.458	0.454	0.465	0.482	0.450	0.406	0.374	0.348	(40)
		taud	2.286	2.202	2.155	2.003	2.132	2.185	2.156	2.104	2.157	2.270	2.311	2.308	(41)
	Ebn at Noon		239	252	257	250	260	261	256	246	244	239	227	226	(42)
		Edh at Noon	27	35	42	53	48	46	47	47	42	33	27	25	(43)
(44) All-Sky Solar Radiation	RadAvg		455	715	1076	1462	1785	2003	2030	1807	1302	846	488	377	(44)
	RadStd		69	102	129	165	155	157	133	149	107	116	68	54	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (55 neighbors)	+1.01	N/A	N/A	N/A	+0.72	+0.72	N/A	-274	+139	N/A

Nomenclature: See separate page